

July 13, 2026 03:51 PM GMT

US Corporate & Securitized Credit | North America

Data Center/AI Financing: What We Have Heard from Investors

AI-related issuance is surging as fixed income markets are proving flexible and companies are consistently willing to broaden financing beyond traditional avenues. We think investors are still adding exposure to AI/data centers in their portfolios, although sentiment is turning incrementally cautious on potential macro/thematic risks that lie ahead. We see a preference for shorter duration and amortizing structures.

Key Takeaways

- Across the different credit markets, we think investors are showing a preference for deals and structures that help offset thematic risks (overbuild, obsolescence, and bottlenecks are the most often mentioned);
- For data center construction debt, thematic risk manifests itself as asset risk at lease renewal; for corporate unsecured debt, thematic risk is inherent to the ongoing investment cycle, especially in the longer run, and therefore credit quality should matter;
- For IG, hyperscalers' funding mix has been the main focus, with companies broadening their investor base overseas and signaling to the market their confidence in the capex cycle and in the monetization opportunities;
- For HY, the debate has been around potential construction delays, evolving structures, and potential refinancing paths;
- In securitized, investors are most focused on supply & demand dynamics. We do not think credit supply has thus far displaced ABS or CMBS issuance, but we're cautious that supply could come in lower-than-expected this year and going forward as other markets offer size and competitive cost of capital for data center companies pursuing rapid growth.

As AI-related debt continues to blur the lines between corporate and securitized credit, we revisit our risk framework definitions with a cross-asset approach and discuss investor feedback on our issuance estimates for the rest of 2026. See more inside:

- [Revisiting Our Risk Framework Definitions](#)
- [Taking Stock of Recent Moves; Reiterate Our Supply Call and U/W IG Tech](#)
- [Investment Grade Credit: Issuance Estimates, Funding Mix, and Index Eligibility](#)
- [High Yield Credit: Evolving Structures, Construction Risk, and Refinancing Paths](#)
- [Securitized Credit: A Supply Drought? Not So Fast](#)

MORGAN STANLEY & CO. LLC

Fernanda Lima

Strategist

Fernanda.Lima@morganstanley.com

+1 212 761-3021

Carolyn L Campbell

Strategist

Carolyn.Campbell@morganstanley.com

+1 212 761-3748

Vishwas Patkar

Strategist

Vishwas.Patkar@morganstanley.com

+1 212 761-8041

Vishwanath Tirupattur

Strategist

Vishwanath.Tirupattur@morganstanley.com

+1 212 761-1043

James Egan

Strategist

James.F.Egan@MorganStanley.com

+1 212 761-4715



See our other related notes:

- [Podcast | Thoughts on the Market: Inside the AI Debt Surge \(19 Jun 2026\)](#)
- [Sunday Start | What's Next in Global Macro: The Convergence of AI and Energy Financing \(14 Jun 2026\)](#)
- [Global Credit Strategy & Securitized Products Research: AI Debt Financing Tracker: Warming Up for a Hot Summer \(9 Jun 2026\)](#)
- [Securitized & Corporate Credit: Data Center Financing: Faster, Broader, Deeper \(26 May 2026\)](#)
- [Podcast | Deep Thoughts on Securitized Markets: Data Center Financing and Value Across Securitized and Corporate Credit | Ep. 360 \(29 May 2026\)](#)



Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

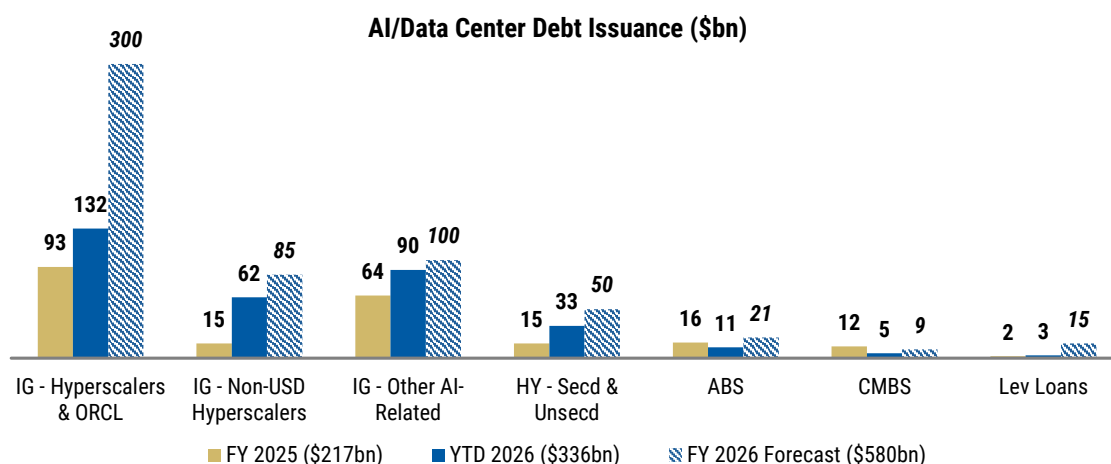
Revisiting Our Risk Framework Definitions

Data-center debt financing remains a critical topic within the credit market, driving our above-consensus issuance forecasts and sector preferences, as well as the overall direction of spreads. In our mid-year outlook from late May, we updated our issuance estimates and spread targets for US credit. We followed this up with an [in-depth report on the outlook for data center debt financing across the corporate and securitized markets](#).

Over the past few weeks, we have discussed our outlook across investors, ranging from macro accounts to core credit investors, and specialists within the corporate and securitized credit worlds. The discussions understandably have spanned a range of topics from hyperscaler capex intensity to market demand capacity, risks around the AI build-out, political backlash, and downside scenarios.

Over the past two weeks, the underperformance of AI-related debt – particularly among hyperscalers – amid a broadening out of issuance since June has been very topical. We address this topic briefly in the [next section](#), and discuss our broader thoughts on the market at the Morgan Stanley Macro Forum on July 13 ([webcast/slides](#)). In this publication, we focus more on insights from our investor conversations and where our views differ from market consensus.

Exhibit 1: We're now at \$336bn of AI-related debt issuance YTD across global credit markets



Source: Bloomberg, Trepp, CreditFlow, PitchBook, Morgan Stanley Research estimates. Data as of 7/10/26.

The details on investor feedback are in the following sections across [investment grade](#), [high yield](#), and [securitized credit](#). However, before we go into the specifics, we clarify some of the terminology that came up in our client meetings and are regularly used in subsequent sections of this report.

The terms below are largely the primary risk vectors associated with the data center construction debt that came up frequency in our investor conversations based on what we are seeing in the HY market (and some in IG): construction risk, tenant risk, and macro/thematic risk. We think these vectors will be significant drivers of relative value across different deals and within asset classes.

- **Construction Risk:** This is the risk associated with the actual build-out of the data

center, i.e., can the developer deliver the data center on time to the tenant without material cost over-runs. Such risk is relatively new to the public corporate credit market, as it is typically underwritten as project finance via real-estate or infrastructure financing channels via construction loans. **This creates an important differentiation between corporate and securitized credit:** IG/HY deals have different degrees of construction risk (brownfield vs. greenfield, contractors involved, construction contracts, overall project complexity, and more). On the other hand, ABS/CMBS deals are fully stabilized assets post construction, which are cash-flowing (with one exception in CMBS).

- **Tenant Risk:** This is the risk associated with the tenant's ability to make lease payments on time and is a reflection of the tenant's creditworthiness. **Once construction is completed, this becomes the key risk, which should be reflected in pricing/risks based on the counterparty's credit quality.** There are some structural enhancements that can help mitigate tenant risk: 1) amortization features – in *corporate credit*, in most cases amortization is sculpted as an annual percentage of principal to maintain a minimum DSCR ratio in line with covenant requirements. In ABS, amortization can be triggered by a) breached DSCR or LTV levels before their anticipated repayment date (ARD) until the condition is cured, or b) if the transaction fails to refinance by the ARD, after which it starts fully amortizing down; in CMBS, there is no amortization mechanism; and 2) in cases in which the tenant has limited or no track record in the credit markets, higher-quality credits have offered backstops and/or lease guarantees (we also refer to those as IG wrappers).
- **Asset Risk: This is the risk associated with the value of the asset itself, i.e., does the ultimate tenant/hyperscaler providing the credit support actually need the data center that has been built.** The inputs associated with a data center facility could be location (proximity to metro markets), workload type (training vs. inference), environmental risks (areas that are subject to natural disasters and/or that can be directly impacted by climate changes), and technology obsolescence (rack power density, cooling and power systems, and other structural components that might impact the facility's ability to support AI hardware upgrades). **Asset risk is already part of the underwriting assessment, but we think it likely becomes a more significant concern in the longer term, when the facilities have to go back to market for a lease renewal.**

We also discuss broader risks that are applicable to both data center construction and corporate debt:

- **Corporate Risk:** In our view, corporate risk is directly linked to the companies' ability to monetize the ongoing investment cycle. Thinking of the investment grade universe, this ties back to hyperscalers being able to get the expected return on their investments in a timely manner; as for the leverage finance complex, it is subject to the companies' ability to establish a successful track record (neoclouds and former bitcoin miners/now data center developers are relatively new to their main business activities, and arguably – more importantly in this context – they are also new to the credit markets). In securitized credit, the data center companies that issue debt are often a secondary consideration to the assets themselves, but the recent [challenges from telco company Extenet](#) demonstrate that sponsor risk

can be very real and cause problems for the bonds even if the underlying assets are still performing.

- **Macro/thematic Risks:** We think risks in this bucket derive from demand for compute and AI adoption, and we now divide these into two opposite camps:
 - **Surplus supply/demand underwhelming:** Risks associated with supply ultimately exceeding demand in the future. It could be driven by overbuild (not only for the final data center capacity, but also across the supply chain as different industries are involved as suppliers), but also by efficiency gains (for compute, power, token usage, and others).
 - **Scarcity:** Among access to power, supply chain bottlenecks, potential federal, state and local government restrictions (such as imposition of moratoria), and NIMBYism, build-out constraints become insurmountable. Capacity limitations drive price increases and users (both the consumer but also enterprises) have to rethink their AI adoption ambitions.

Taking Stock of Recent Moves; Reiterate Our Supply Call and U/W IG Tech

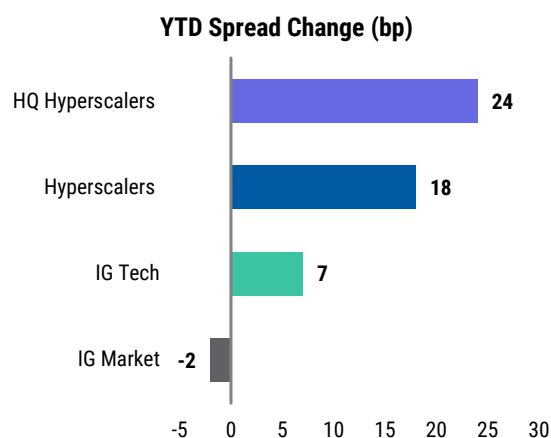
The dynamic between supply and spreads for hyperscalers is a key theme in the IG market. Prior to June, investors had grown comfortable with the view that AI-related funding needs would largely come from hyperscalers and ORCL, while these issuers also sought to improve transparency and establish a more regular issue cadence in response to investor feedback. **We think funding mix diversification via non-USD debt and equity issuance (Exhibit 7) also contributed to investor perception of manageable supply pressures in the USD market.**

Since June, however, we think the "surprise" element has weighed on spreads (Exhibit 2). In the past ~40 days, the market saw sizable issuance from other mega caps, hyperscaler issuance ahead of earnings, and most recently an [ORCL ratings downgrade by S&P](#). We think the latter is likely to bring fundamental/quality risks to the forefront.

As a result, consistent with our U/W view, we are seeing spread decompression across the AI complex vs. the market (Exhibit 3), while other sectors have held up well. We think this is largely a function of strong demand, meaning investors likely do not need to reduce exposure to other sectors to accommodate hyperscalers. **To the extent that rates stay high, insurance and overseas appetite could mean that AI decompression persists but doesn't spill over to the overall IG index. However, if rates go lower (our baseline view), we think demand may also slow and lead to a broader repricing in the market, particularly if supply remains elevated.**

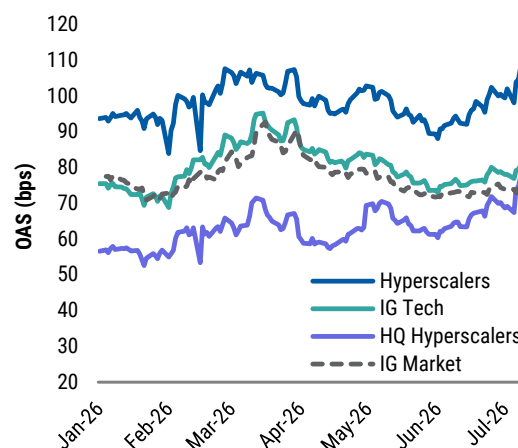
Finally, the large transactions issued since June are trading wider than their new-issue pricing, a sign of sentiment shifting. We continue to forecast an above-consensus total IG supply of \$2.25tn (vs. ~\$1.33tn YTD) and expect spreads to underperform other fixed income markets. We remain U/W Tech within the IG market, due to our expectation for continued supply and potential credit quality headwinds.

Exhibit 2: IG index is ~2bp tighter YTD; HQ hyperscalers are now ~24bp wider YTD



Source: Bloomberg, Morgan Stanley Research. Data as of 7/10/2026 EoD.

Exhibit 3: High quality hyperscalers are now also wider to the broader IG market



Source: Bloomberg, Morgan Stanley Research. Data as of 7/10/2026 EoD.

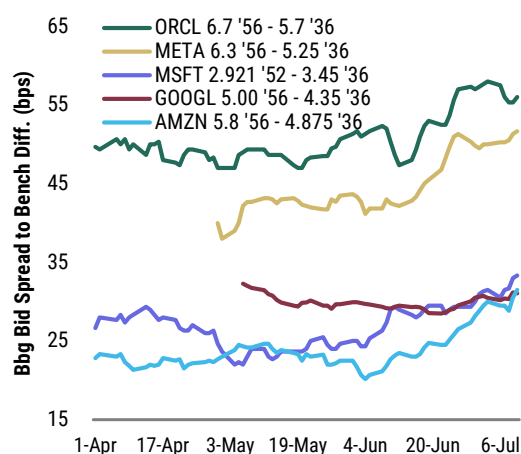
Investment Grade Credit: Issuance Estimates, Funding Mix, and Index Eligibility

The most frequently asked questions across IG revolved around our FY26 issuance estimates, how supply is impacting spreads (reinforced by [recent volatility](#)), equity funding vs. debt funding, and non-USD issuance out of the hyperscalers. With respect to data center construction bonds, market capacity and pipeline for large structured IG "private-style" deals also came up. Further, for such deals, since bonds are 144A for life, index eligibility and deal structure/marketability were also topics of questions.

Until recently, we think investors were more comfortable underwriting corporate risk over macro/thematic risk. This translated into demand for high-quality paper even if supply from such issuers is elevated, at the expense of lower-rated companies where supply might be muted. However, future credit profiles are very tied to the AI demand thematic. When it comes to the hyperscalers and Oracle, there is a fair and valid argument to be made that in the longer run there is a much higher degree of correlation between corporate and thematic risks, even more so for companies that don't benefit from ample business diversification.

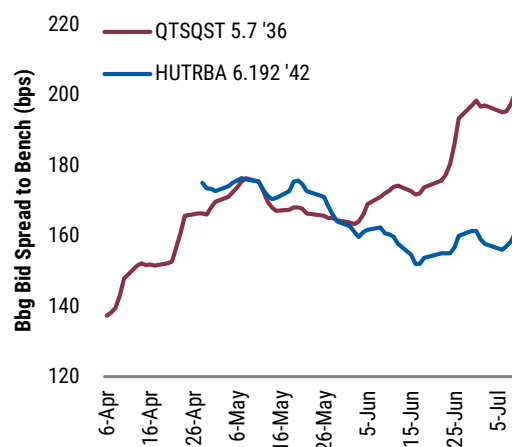
There is broad awareness that actual leverage for hyperscalers and Oracle is increasing through the different off-balance sheet channels (e.g., purchase commitments, lease commitments, backstops and guarantees, asset-backed liabilities at SPVs, and more). However, while we await for investment returns to start playing out, credit quality, business diversification, financial policy, and other attributes matter for investment decisions, and we can already see differentiation across the various names ([Exhibit 4](#)). We think that ORCL's earlier-than-expected ratings downgrade could shift the debate beyond just supply to more fundamental tiering across the different players.

Exhibit 4: 10s30s curves for hyperscalers and ORCL



Source: Morgan Stanley Research, Bloomberg. Note: Updated 7/12/26; levels based on a 3-day moving average.

Exhibit 5: QTSQST trades ~40bp wide to HUTRBA



Source: Morgan Stanley Research, Bloomberg. Note: Updated 7/12/26; levels based on a 3-day moving average.

We caveat that the sample set of data center construction bonds in IG is limited (see [Exhibit 13](#)). Within this sample, though, we think investors favor the deals with lower

asset risk and higher-quality tenants. Tenant risk is ultimately corporate risk, and therefore investors are more familiar with how to underwrite this type of risk. Asset risk arises at the time of lease renewal, in our view, and becomes more acute when there is no amortization during the life of the lease. Looking at the relationship between Hut 8 and QTS bonds ([Exhibit 5](#)), a key takeaway that stands out is that construction risk and the issuer's limited track record as a developer can be offset with a construction guarantee and other structural enhancements, while asset risk remains a concern even when the tenant is a high-quality counterpart.

- For context, Hut 8's notes are a construction-phase financing, with construction risk mitigated by a design-build contract guaranteed by Jacobs Engineering. Importantly, the debt is structured to fully amortize within the 15-year lease term (which matches the maturity of the notes). QTS financing, by contrast, is for a fully-built and operational asset leased to MSFT for 20 years. The debt isn't amortizing and matures in 10 years. Refinancing risk isn't the only consideration, though, as even at lease renewal in 2046, Moody's estimates that ~50-65% of principal would still be outstanding.

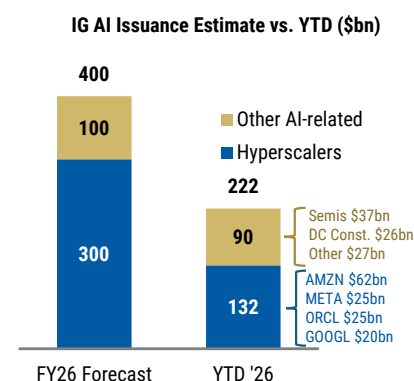
How is YTD issuance tracking vs. our estimates?

We're forecasting \$350-400bn of issuance within IG. We break down our issuance estimate into two categories: 1) \$250-300bn from the HQ hyperscalers + Oracle; and 2) ~\$100bn from other AI-related debt, which so far includes issuance from semiconductor companies, data center construction projects (both privately and publicly syndicated), data center REITs, and other companies directly related to the AI capex cycle.

As of July 10, **AI-related debt issuance in the IG market totaled ~\$222bn of YTD (or \$257bn when including the AVGO chip financing transaction).** To achieve our \$400bn estimate for AI-related issuance in the IG market pace in 2H would have to be at least the same as in 1H, which we think is quite possible when looking at current 2027 capex estimates.

We see some merit in investor feedback that **USD issuance from hyperscalers and ORCL (\$132bn YTD) could end the year below \$250-300bn given the amount of non-USD issuance (\$62bn YTD)**, and that other AI-related debt could exceed \$100bn (vs. ~\$90bn YTD currently). However, given the pace at which capex needs are rising among the five main companies (+54% in 2027; see [Exhibit 11](#)), we maintain our current estimates and expect issuance to remain elevated in 2H, particularly as issuance capacity in other currencies is likely to have been exhausted in the near-term.

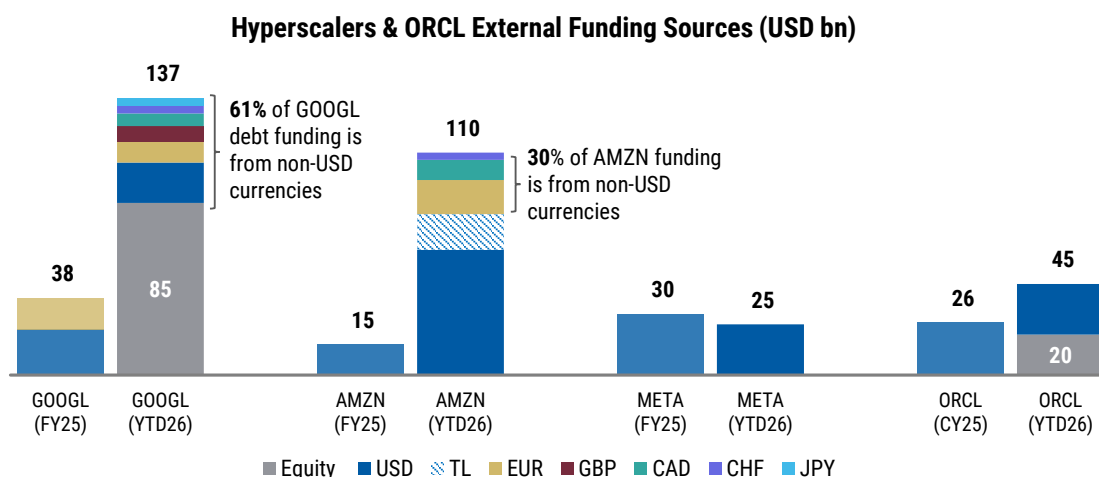
Exhibit 6: USD IG issuance is at \$222bn YTD



Source: Bloomberg, company filings, Morgan Stanley Research.

We expect the hyperscalers to continue to explore different financing avenues, with cost of capital being a secondary consideration vs. the opportunity of broadening the investor base ([Exhibit 7](#)). Since 2025, companies have issued debt in five non-USD currencies (EUR, GBP, CHF, CAD, and JPY), Amazon announced a [\\$17.5bn delayed drawn TL](#) (DDTL) with a 3-year tenor and issued at S+75 (based on its current ratings), and both ORCL and GOOGL announced equity issuance transactions this year ([Exhibit 8](#)).

Exhibit 7: Hyperscalers & ORCL external funding mix now includes equity and six different currencies



Source: Bloomberg, Company filings, Morgan Stanley Research.

What does equity issuance means for our supply estimates?

While we don't have specific information from the companies around the incentives behind equity issuance, we think fundamentally equity issuance can be a funding source in the following scenarios:

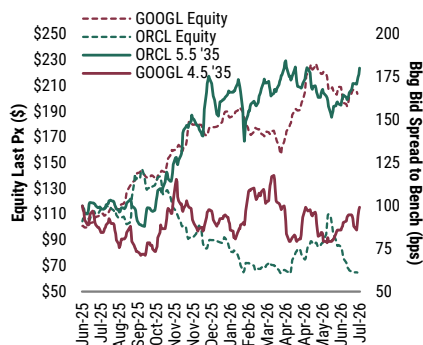
1. A solution for companies that can't access the credit markets. We acknowledge that common equity valuation might not be compelling in such cases, but we're thinking of preferreds and other instruments that include monetary incentives for investors that agree to participate. We don't think any of the hyperscalers fall in this bucket.
2. A way to prevent leverage from further increasing and therefore continue to access the credit markets. We place ORCL's announced equity issuance in this bucket.
3. An indication of commitment to a broad and sizeable investment cycle that requires significant capex for a number of years.

We also note that beyond AI-related capex, funding is also needed for acquisitions and other investments. This is particularly relevant for Alphabet, in the context of Waymo's latest funding round (\$16bn, with the vast majority funded by Alphabet - see [pg. 39 of the 1Q26 10-Q](#)), the Intersect acquisition for ~\$6bn (see [pg. 25 of 1Q26 10-Q](#)), and the Wiz acquisition for \$29.5bn ([pg. 24 of the 1Q26 10-Q](#)).

Our best guess is that the hyperscalers' comments on potential equity funding align with point #3 above, i.e., a large and broadening investment cycle that is unlikely to slow down any time soon, and hence companies may seek to have access to a broad range of funding

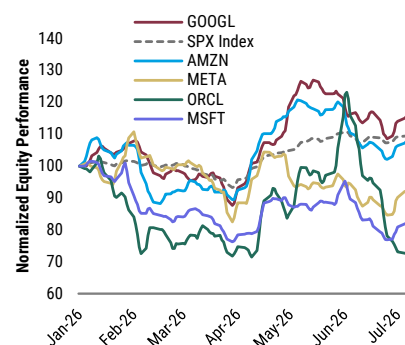
channels and preserve financing flexibility across markets. **In other words, we don't see equity funding being a replacement for debt funding.** Therefore, we aren't changing our supply estimates for now.

Exhibit 8: Bond and equity performance for GOOGL and ORCL



Source: Bloomberg, Morgan Stanley Research. Note: Updated 7/12/26, using a 3-day moving average.

Exhibit 9: Hyperscalers & ORCL equity performance YTD



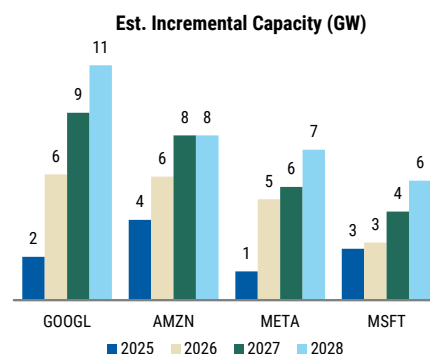
Source: Bloomberg, Morgan Stanley Research. Note: Updated 7/12/26, using a 3-day moving average.

Is there upward bias to 2027 capex estimates?

Yes, in our view. Thinking about the broader investment cycle, we believe the largest YoY increase still takes place in 2026. However, [Morgan Stanley equity analysts' recently revised estimates](#) already point to a 54% increase in 2027 ([Exhibit 11](#) - we're looking at the high quality hyperscalers and ORCL for now). We outline a few reasons why:

1. Numerous incentives for more forward-building ([Exhibit 10](#)): According to our [equity colleagues](#), the numerous bottlenecks associated with data center constructions have extended the timeline from groundbreaking to operational launch to as much as three years. To the extent that permitting and zoning for data center construction continues gets stricter in response to local backlash, we think there is an incentive for hyperscalers to kick off projects for forward capacity sooner rather than later. According to Morgan Stanley's US public policy strategist Ariana Salvatore, the key risk to monitor is that the [local pushback against data centers becomes a federal issue](#) – we think sensitivity to this risk likely increases as we start to approach the next presidential election in 2028.

Exhibit 10: High quality hyperscalers to add ~27 GW of add'l data center capacity in 2027

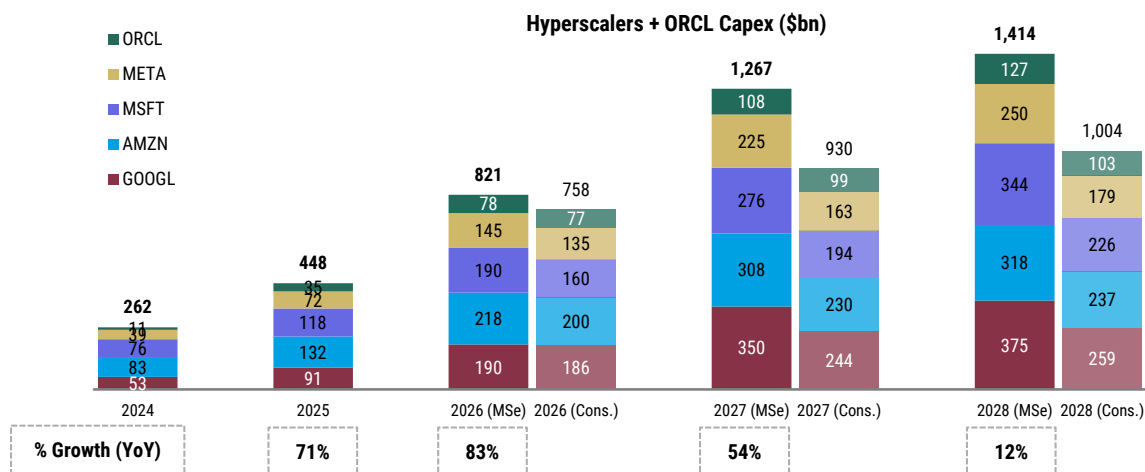


Source: Morgan Stanley Research. See the following note:

2. Racks and servers are the biggest cost drivers for data centers. So far, public markets have seen more transactions to finance the construction of data center shells. Given the possibility of leasing such structures, we've seen a number of first-time issuers in both the IG and HY markets out of companies that aspire to be data center developers. As the shells near completion, we expect to see more financing for chips/compute, which actually

represent ~60% of the cost to build a data center (Exhibit 12). The decision to own or lease chips will influence the financing framework (on vs. off balance sheet). Increasing memory costs, as well as newer generations of chips, could also result in higher costs per GW.

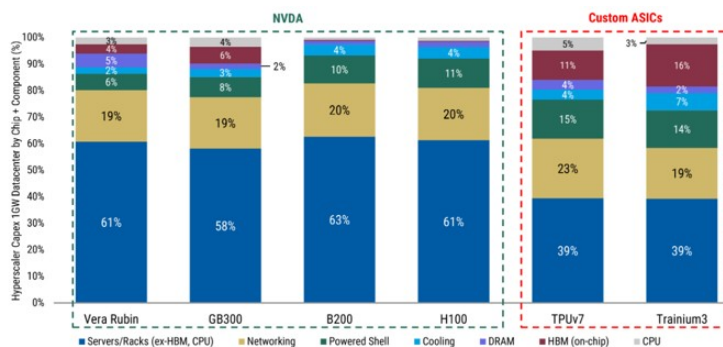
Exhibit 11: Current estimates point to a 54% increase for total capex from hyperscalers and ORCL in 2027



Source: Company filings, Bloomberg & Visible Alpha for consensus, Morgan Stanley Equity Research. Morgan Stanley Research estimates for AMZN/GOOGL/META (covered by Brian Nowak) are from the following note: [Internet: \\$1.4trln of Capex, 4X in Compute Capacity, and AI Revenue Streams to Watch into '28](#) (12 Jul 2026). Morgan Stanley Research estimates are as of 5/4/26 for MSFT (covered by Josh Baer) and as of 4/23/26 for ORCL (covered by Sanjit Singh).

3. Volume vs. inflation. As we think of future capex growth, a frequent question is the split between volume growth and inflation/price increases, especially as we get more data indicating shortages and longer lead times for electrical and other types of equipment. Based on a 2025 report from [Turner & Townsend](#) that discusses data center construction costs, there's a 5.5% increase in the cost per watt of building a traditional data center. There is also a 7-10% cost premium for liquid-cooled data centers in the U.S.. Finally, 60% of survey respondents expect construction costs to increase by 5 to 15% in 2026, with 21% expecting >15% of inflation in the year.

Exhibit 12: Compute hardware, including servers/racks, HBM and CPU represent ~60% of the cost/GW to build a data center



Source: Morgan Stanley Research; screenshot of slide 26 from the following note: [Internet: How Much Capacity Will \\$2 Trillion of Hyperscaler Capex Bring By '27?](#) (17 May 2026).

Can index eligibility and offering format impact performance going forward?

As the IG construction bonds were issued as 144A-for-life with no registration rights, **investors have been asking how the lack of index eligibility could impact their trading levels going forward.** Benchmark weights and passive-fund demand can certainly impact holdings, liquidity, and spreads. To the extent it does impact performance of existing bonds ([Exhibit 13](#)), we think issuers are faced with two options: 1) offer a premium for index inclusion; or 2) consider moving to an unrestricted CUSIP, as other issuers have chosen to do. On the latter, we think it is worth circling back to our analysis from a few years ago on [NFLX bonds after they got IG ratings](#). We learned back then that **144A-for-life bonds may, in certain circumstances, be converted into freely transferable, unrestricted global securities (which are eligible for index inclusion).**

[SIFMA guidance](#) provides that, for already-outstanding 144A-for-life securities, an issuer may elect to implement a variant of the Rule 144 procedures to promote secondary-market liquidity or potential index eligibility. **At any time after the securities have been outstanding for at least one year, assuming the conditions for free transferability by non-affiliates are satisfied and the governing documents do not prohibit the procedure, the issuer may deliver a free transferability certificate to the trustee.** Once the certificate is delivered, the restricted securities can be moved into a new unrestricted global note with a new unrestricted CUSIP. Bloomberg would then be notified and update its systems to reflect the unrestricted CUSIP. This may support eligibility for index inclusion, though eligibility is not automatic and remains subject to the relevant index rules and other bond-specific criteria.

Exhibit 13: There's ~\$53bn outstanding in data center construction bonds that currently are not index eligible

Issue	Amt. Out. (\$mn)	Market	Issued with Reg. Rights?	Ratings (S&P/Moody's/ Fitch)			Issue Sprd to Bench (bps)	Current Sprd to Bench (bps)	Sprd Δ Since Issuance (bps)
RPLDCI 6.581 '49	27,294	PRIV PLACEMENT	N	A+	NR	NR	225	211	(15)
RDMDCH 7.500 '45	14,000	PRIV PLACEMENT	N	NR	Baa3	BBB	329	348	19
HUTRBA 6.192 '42	3,250	PRIV PLACEMENT	N	(P)BBB-	NR	BBB-	185	161	(24)
HUTBPA 6.129 '42	4,250	PRIV PLACEMENT	N	NR	Baa2	NR	165	165	0
QTSQST 5.700 '36	4,600	PRIV PLACEMENT	N	NR	Baa2	NR	138	209	72

Source: Morgan Stanley Research, Bloomberg. Note: Spreads as of 7/10/2026, changes calculated using unrounded spreads.

Finally, the depth and capacity of the IG private placement 4(a) (2) format has come up in our investor conversations given the recent \$35bn deal for Broadcom (see this [recent note](#) from credit analyst Lindsay Tyler for additional details). Issuers may choose 4(a)(2) format because it allows them to raise capital privately, quickly, and with greater control over disclosure, investor selection, and documentation. This can be especially attractive for sponsor-backed issuers, project-finance SPVs, or highly bespoke financing structures, such as this one was (including multiple tranches with delayed draw features, per [The Financial Times](#)). With no public offering aspect involved, the deal can be privately negotiated with large sophisticated investors rather than built for public-style distribution. Also, according to [The Financial Times](#), the debt could become tradeable in the institution-only 144A bond market.

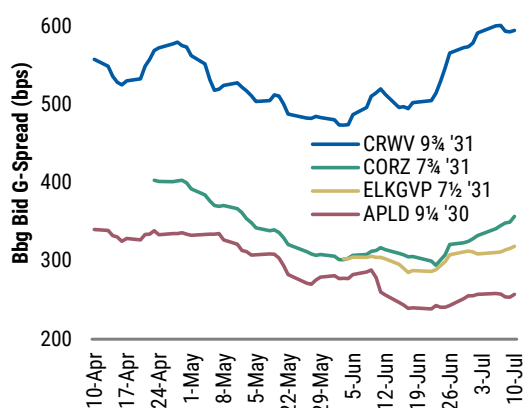
High Yield Credit: Evolving Structures, Construction Risk, and Refinancing Paths

Our conversations with HY investors had a slightly different flavor. While supply remained a key topic, investors concentrated more on understanding fundamental risk factors, potential take-out scenarios and refinancing paths, as well as potential rating agency upgrades once construction is complete.

Investors generally agree with our issuance estimates (\$50bn for HY/\$15bn for leveraged loans), and therefore **questions were centered around key risks, how structures are changing given growing supply, and potential refinancing paths**. The HY deals are 5NC2, and per our conversations, investors think bonds could be called on the first call date, as issuers would like to obtain lower cost of capital (given the deals would no longer have construction risk).

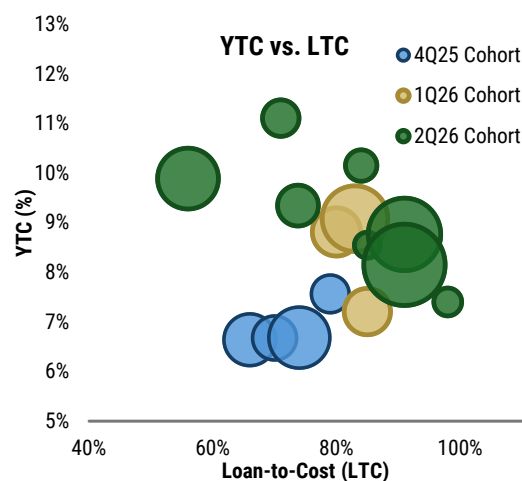
Construction risk over unsecured corporate risk in HY (Exhibit 14): Given the shorter tenor of bonds (and loans) in the HY market vs. the IG market, we think the debate for HY is, for now, mostly focused on construction risk. The preference for taking risk at the project level vs. the corporate level makes sense, in our view, given corporate issuers (and HY-rated tenants) are newer companies (vs. the IG-rated hyperscalers) with limited track record and are still growing into their capital structure.

Exhibit 14: CRWV unsec'd bonds vs. project finance-like DC construction bonds in which CRWV is the tenant



Source: Bloomberg, Morgan Stanley Research. Note: Updated 7/12/26, using a 3-day moving average.

Exhibit 15: LTC ratios have increased since 4Q25, with more dispersion in the 2Q26 cohort



Source: Morgan Stanley Research, Bloomberg, Moody's, S&P, and Fitch. Note: Updated 7/12/26.

Investors are building frameworks to evaluate the different assets and deal structures.

We received a number of questions on construction delays, location (from environmental risks to distance vs. big markets given training vs. inference usage), design, and construction partners as well as construction contracts.

While construction risk remains a key topic in most conversations, **investors generally agree with our view that construction delays could result in limited sell-off, and**

ultimately in some dispersion based on developer's track record, but are unlikely to result in cancellations from tenants given constrained access to power, challenges related to permitting & zoning, and increasing supply-chain inflation and bottlenecks.

How are the deal structures changing over time?

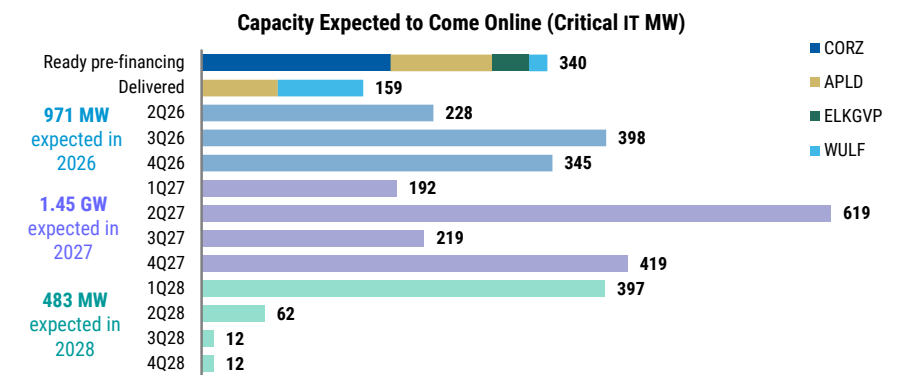
Current levels indicate a preference for deals that offer faster deleveraging, and we got the same feedback from investors during our conversations.

- **Loan-to-cost (LTC) has increased** (see [Exhibit 15](#)), especially for deals that have either a high-quality hyperscalers as tenant or an IG wrapper (by wrapper we mean either a full lease guarantee or a backstop);
- **Amortization is starting later on** (closer to the refinancing window) and **has decreased** (lower % of principal per year, with amortization payments often sculpted to meet minimum debt service coverage ratio requirements over time);
- **Termination rights have evolved to benefit the developers**, with termination rights for construction delays either fully removed or moved farther away in time (more than one year vs. six months initially);
- **Cash waterfall post debt service (interest and amortization) has gotten looser** (some of the 2025 deals had mandatory ECF offers in the waterfall, while the most recent deals will allow for any use of excess cash flow that isn't prohibited by the indenture);
- **Construction projects are increasingly shifting to multi-phase/multi-building, with numerous staggered delivery dates** and smaller data halls (which allow construction to move faster once developer and tenant agree on the design and other aspects of the first data hall).
- **Cost per MW has increased**, as well as the timeline between deal announcement and first delivery date, and between first and last delivery date. This is expected, in our view, due to inflation and supply-chain bottlenecks.

Will we see construction delays?

Our initial expectation is that yes, we could see some delays. We estimate that >740 MW of critical IT capacity could come online during 2H26 ([Exhibit 16](#)), and we think that anchored investor expectations around potential delays will be key to support current trading levels. Other than brief mentions on issues, such as small delays due to tenant design changes, or substation construction items being difficult to source, we haven't heard public commentary from issuers about potential construction delays to date.

Exhibit 16: The first wave of deliveries starts in the fall, but most of the capacity from HY deals is expected to come online in 2027



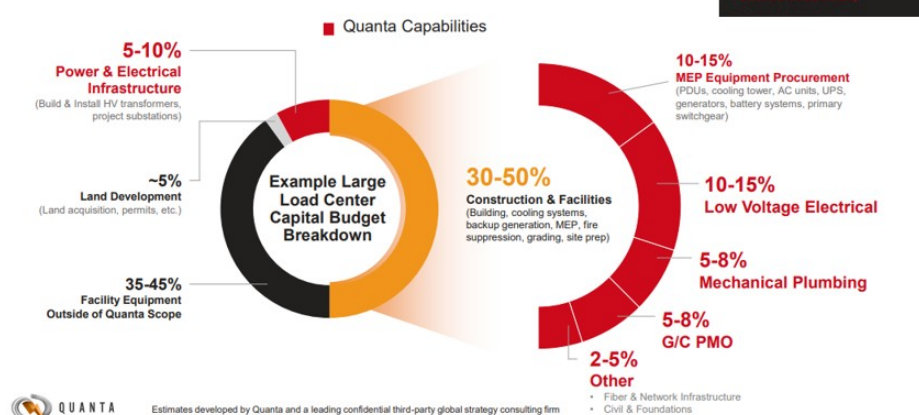
Source: Morgan Stanley Research, Bloomberg, Moody's, S&P, and Fitch. Data as of July 2026.

We present some of the potential drivers of construction delays below. None of these are new to data center construction (an [FTI Consulting report](#) from October 2024 already flagged many of the same reasons discussed in this note and more), and we expect to get more data points on what is driving potential delays, if there are any. We expect some updates as we go into 2Q26 earnings.

- 1. Compressed timelines** (construction is fast-paced, with timelines between 12-18 months in most cases; \$1bn construction projects could take 3-5 years depending on the complexity);
- 2. As much as the construction of the shells isn't extremely complex, the associated systems – power, cooling, and more – are very complex, which creates interface risk;**
- 3. There could be design changes through different stages of the project,** depending on when/how involved the tenant is since inception (we've seen this happening recently, but note that this isn't new to data center construction, and companies such as Digital Realty and Switch have flagged similar issues in their regulatory filings in the past);
- 4. Supply-chain bottlenecks and labor constraints** are worsening (see some key stats on the [Sunday Start from June 14](#));
- 5. Even with the existing fixed-price construction contracts for some of the projects, developers are keeping a relevant share of responsibility for construction to be done on time, and these are companies with limited track records so far.** Based on analysis by Quanta Services ([Exhibit 17](#)) , an IG-rated construction company involved in the space, 35-45% of facility equipment required for data center construction sits outside of their scope.

Exhibit 17: Data center spend breakdown per responsible counterparty: developer vs. construction partner

Positioned To Capture The Majority Of Craft-Led Spend On A Data Center



Source: Screenshot of slide 24 from [Quanta Services June 2026 Investor Presentation](#).

What are the potential refinancing paths for the HY data center construction bonds?

Assuming construction completes on schedule and the AI-ecosystem fundamentals remain strong, we think HY bonds can be refinanced with a lower cost of capital and have multiple potential avenues for refinancing. As a result, elevated supply in HY might not be a multi-year trend (unlike IG). Investors have asked us whether some issuers could look to the ABS or CMBS market to refinance some of these projects. From the corporate credit side, we think there are a few important factors to keep in mind:

The first call date for these bonds is generally after the projects' final delivery date (not accounting for potential delays – Exhibit 18). For now, the 2-year non-call period covers the estimated construction period of the majority of the deals. The ability to refinance, in our view, is directly impacted by the successful completion of construction. Not only will it lower the cost of capital, but the possibility of choosing how to refinance outstanding debt will also only be available for developers that can deliver against promised timelines.

Can the ABS market provide cheaper refinancing options vs. HY? Based on the actions of HY telecom companies since 2022, we think this is an important consideration, but the answer here isn't a straight-forward "yes", in our view. ABS could provide cheaper financing per turn of leverage, given its structural enhancements and protections, and this becomes important as LTC ratios go up and deals present a slower deleveraging profile. WAVG cost of capital in the ABS and CMBS market this year is around 5.5-6.5%, depending on the type of structure and duration of the asset (Exhibit 21).

The loan market is another path to consider. We haven't seen too much activity in terms of data center construction loans in the BSL market. However, to the extent that call features remain attractive to issuers even after construction gets completed, we think issuers could look into refinancing the HY bonds with loans: call protections are usually shorter (<1 year) and facilities can be repriced if there is further opportunity to lower the cost of capital as issuers continue to build their track record. We expect the rates path to

Exhibit 18: For 13 out 16 deals the first call date is after the expected final delivery date (assuming no delays)

Issue	First Call Date	Final Delivery Date
YNDRDC 6.875 '31	6/30/2028	Oct-28
ELNFOR 7.000 '31	6/15/2028	Jun-27
STNGRY 6.000 '31	6/15/2028	Apr-27
ELKGVP 7.500 '31	6/15/2028	Jan-27
SECMOS 8.875 '31	5/1/2028	Jun-27
TRACTD 6.500 '31	5/1/2028	Apr-28
CORZ 7.750 '31	5/15/2028	Jun-27
EDGCOM 7.500 '31	4/30/2028	Dec-27
MERIDI 6.250 '31	4/30/2028	Feb-28
PFORGE 6.750 '31	3/15/2028	Jun-27
TRACTC 5.875 '31	3/1/2028	Jun-28
BLKPRL 6.125 '31	2/15/2028	Feb-27
FLASHC 7.250 '30	12/31/2027	Sep-26
APLD 9.250 '30	12/15/2027	Oct-26
CIFR 7.125 '30	11/15/2027	Sep-26
WULF 7.750 '30	10/15/2027	Dec-26

Source: Morgan Stanley Research, Bloomberg, Moody's, S&P, and Fitch. Notes: Blue denotes deals with first call date outside of final delivery dates, yellow denotes deals with first call date before the final delivery date. Gray denotes deals in which first call date and final delivery date are very close.

also influence a potential issuer preference for loans given their floating aspect.

Issuers might have different aspirations. We've now seen deals from private project finance-style SPVs, and others that are more likely viewed as corporate/hybrid data center construction projects. We think issuers of the latter might be more inclined to refinance in the corporate credit market to build a track record as repeat issuers, which could allow them to issue debt with fewer structural enhancements in the future.

Will the securitized credit market be able to absorb a refinancing wave?

From the securitized credit perspective, we think the market could be open to seeing those deals come through. In our view, HY deals offer two important things that the ABS and CMBS market currently lack: 1) significantly more information and transparency about the tenant and lease terms; and 2) partial amortization. In order to crowd in more demand from securitized credit investors – which we think will be necessary to take down supply – we think better transparency and deleveraging structures could be necessary (see our note [Something's Gotta Give](#) for more on this dynamic).

That said, the data centers going through the HY market look very different from those in ABS/CMBS. Data center deals with a high exposure to neoclouds have priced wider than those backed by hyperscalers to date, and none have been issued by a crypto company. The vast majority of deals are also multi-tenant and multi-campus. Therefore, a HY refinancing wave into ABS/CMBS looks like a compromise, in our view: a different asset risk profile in exchange for better transparency and perhaps more deleveraging.

From a size perspective, data center ABS is just a small fraction of the overall market – about 10% of commercial ABS and 4% of the overall market, with around \$45bn outstanding. The data center SASB market is just about 7% of the overall SASB market or around 4% of all CMBS, with over \$20bn outstanding. We think that leaves plenty of room to grow, with a number of investors still participating in limited size or have yet to enter the market. There also seems to be plenty of money that still needs to be put to work, and so we think demand will be able to grow alongside the supply.

We forecast around \$130bn of remaining issuance in ABS and CMBS from mid-2026 through 2028, with our forecast built upon incremental leased data center capacity in North America and a gradual pick-up in the securitization rate as the market matures; this leased capacity does not distinguish between operators, and naturally our forecast has some single-tenant, single-campus AI training/inference leased capacity built into it.

Can bond ratings be upgraded once construction is complete?

The general answer is yes, but in some cases rating agencies would like to see more than just construction completion for an upgrade. We put together a ratings heat map in [Exhibit 19](#): green denotes upgrade potential once construction is completed, yellow denotes that construction completion by itself likely won't suffice for an upgrade.

We couldn't find language in the reports that clearly indicate a path to a rating upgrade of more than one notch for now. Moody's reports have commentary on potential positive ratings momentum for most of the issuers, specifying that it would require a proven track record from these companies as data center developers. Finally, some deals could be impacted by a change in the tenant's rating (CORZ, EDGCOM, APLD, SECMOS, ELKGVP,

and ELNFOR).

Exhibit 19: We compiled the language from rating agency reports on upgrade thresholds/milestones for the HY data center deals

Issue	Current Ratings & Outlooks			Language from Reports on Rating's Upgrade		
	S&P	Moody's	Fitch	S&P	Moody's	Fitch
YNDRDC 6.875 '31	NR NR	Ba3 STABLE	BB(EXP) STABLE	N.R.	• Timely data center completion and co. builds operational track record in the U.S. U/g would also require maintaining DSCR > 1.1x by 2029 (~1yr after stabilization). (6/29/26)	• Upgrade requires sustained operating and financial performance at/above Fitch case, w/ PLCR >1.35x. (6/23/26)
ELNFOR 7.000 '31	BB- POS	NR NR	BB- POS	• Construction completion alone not enough. Could u/g if APLD is u/g or stand-alone FFO/Debt >9% and EBITDA interest coverage >1.75x, above S&P's base case. (6/9/26)	N.R.	• U/g possible if CRWV improves above 'BB-' coupled w/ on-track completing progress. (7/2/26)
STNGRY 6.000 '31	NR NR	Ba2 STABLE	BB- STABLE	N.R.	• Timely data center completion and operating cash flow sufficient to meet debt amortization. Other factors could add positive rating momentum. (6/12/26)	• Positive action unlikely given potential additional project debt allowance and construction-completion risk. (6/24/26)
ELKGVP 7.500 '31	BB- STABLE	Ba3 STABLE	BB- POS	• Upgrade if CRWV credit quality rises above the project rating or a replacement lease is secured at better pricing than assumed. (6/23/26)	• Upgrade if tenant credit quality improves, data halls are fully completed, and operations meet forecast expectations. (6/5/26)	• Upgrade if CRWV is u/g, provided PLCR remains >1.05x. Timely completion of power infrastructure by ComEd plus Data Halls 4-6 finish commissioning on time. (6/25/26)
SECMOS 8.875 '31	P)BB+ NEG	NR NR	BB- STABLE	• Rating is capped by the lease guarantor; near-term upside is outlook to stable if SoftBank outlook is revised, assuming project performance stays on plan (4/30/26).	N.R.	• Upgrade requires sustained operating and financial performance at/above Fitch case, PLCR >1.05x at refinancing, and tenant credit quality commensurate w/ ratings. (6/22/26)
TRACTD 6.500 '31	(P)BB STABLE	Ba1 STABLE	BB STABLE	• U/g unlikely until construction and debt refinancing are done, plus track record showing add'l debt/events do not weaken credit quality. (5/1/26)	• Upgrade if construction stays on time/budget, data halls meet service levels, and leverage/coverage metrics are at least in line with expectations. (5/8/26)	• Upgrade if all tranches are commissioned in line w/ lease terms, coupled w/ sustained operating/financial performance, with DSCR and PLCR >1.10x. (5/21/26)
CORZ 7.750 '31	BB POS	NR NR	BB- STABLE	• U/g after construction if project cash flow fully services debt, with FFO/Debt >=10% and EBITDA interest coverage >1.8x; earlier if CRWV is u/g to BB-. (4/22/26)	N.R.	• Positive action unlikely given completion/operating risks and a potential additional project with unknown lease term, completion and op. risk profile. (5/28/26)
EDGCOM 7.500 '31	BB+ STABLE	NR NR	BB- STABLE	• After completion, u/g if FFO/Debt >9% and EBITDA interest coverage >1.75x are sustained. Could also u/g if parent EDC Venture is u/g (unlikely near term). (4/20/26)	N.R.	• Upgrade if CRWV is upgraded above 'BB-' and project maintains PLCR >1.15x, along with Fitch's assessment that the likelihood of any potential sanctions event risk relating to the Alibaba lease is extremely low. (6/18/26)
MERIDI 6.250 '31	BB- STABLE	Ba2 STABLE	BB STABLE	• Upgrade unlikely, but possible if project achieves DSCR >1.05x under the Google step-in scenario. (6/12/26)	• Timely data center completion and operating cash flow sufficient to meet debt amortization. Other factors could add positive rating momentum. (5/8/26)	• Upgrade if all tranches are commissioned in line w/ lease terms and operating/financial performance is sustained, with DSCR >1.15x. (5/26/26)
PFORGE 6.750 '31	NR NR	Ba3 STABLE	BB- STABLE	N.R.	• Upgrade possible if data center construction finishes on time and operating cash flow covers debt amortization. Other factors could add positive rating momentum. (3/6/26)	• Upgrade up to the initial maturity of the notes is unlikely due to low DSCR in the initial years. (3/27/26)
TRACTC 5.875 '31	P)BB+ STABLE	Ba1 STABLE	BB STABLE	• U/g unlikely until construction and debt refinancing are done, plus track record of prudent financial mgmt. showing added debt/other events do not weaken credit quality. (5/4/26)	• Upgrade if construction stays on time/budget, data halls meet service levels, and leverage/coverage metrics are at least in line with expectations. (2/18/26)	• Upgrade if all tranches are commissioned in line w/ lease terms and operating/financial performance is sustained, with DSCR >1.18x. (3/23/26)
BLKPRL 6.125 '31	NR NR	Ba2 STABLE	BB- STABLE	N.R.	• Timely data center completion and operating cash flow sufficient to meet debt amortization. Other factors could add positive rating momentum. (2/11/26)	• Upward rating action is unlikely due to the risk associated with potential additional project debt of unknown terms, completion and operating risk profiles. (2/19/26)
FLASHC 7.250 '30	NR NR	Ba3 STABLE	NR NR	N.R.	• Timely data center completion and operating cash flow sufficient to meet debt amortization. Other factors could add positive rating momentum. (12/21/25)	N.R.
APLD 9.250 '30	BB- POS	NR NR	BB- STABLE	• One-notch u/g possible when ELN-3 lease starts, other projects stay on schedule, FFO/Debt >9% and EBITDA interest cov. >1.75x. More high-quality leases could also support upside. (11/10/25)	N.R.	• U/g possible if CRWV goes > 'BB-' and expansion plans do not add completion or counterparty risk. (12/19/25)
CIFR 7.125 '30	NR NR	Ba3 STABLE	BB- STABLE	N.R.	• Timely data center completion and operating cash flow sufficient to meet debt amortization. Other factors could add positive rating momentum. (11/10/25)	• Upgrade if site commissioning is satisfactory and operating/financial performance is sustained, with DSCR >1.2x. (12/10/25)
WULF 7.750 '30	BB POS	Ba2 STABLE	BB STABLE	• Complete La Lupa and Akela on time, increasing confidence that 2027 FFO/Debt >9% and EBITDA interest coverage >1.75x. (10/14/25)	• Timely data center completion, operating cash flow covering debt amortization, proven landlord/operator track record, and DSCR (incl. amortization) >1.2x. (10/21/25)	• Upgrade if CB-3, CB-4, and CB-5 commissioning is satisfactory and performance is sustained, with DSCR >1.2x. (12/4/25)

Source: Morgan Stanley Research, S&P, Moody's, and Fitch. Notes: Dates of each report are included in each of the cells. For Fitch, PLCR stands for project life coverage ratio.

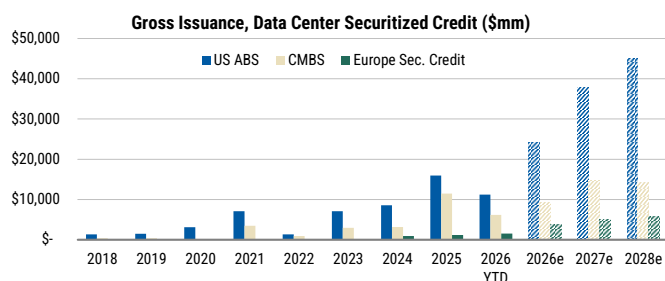
Securitized Credit: A Supply Drought? Not So Fast

In securitized credit, the biggest questions from investors have centered around supply and demand dynamics. With supply quieter in Q2 than in Q1 (and investors were really kept waiting til the end of the quarter), and indeed quieter than the market expected, we think there is a focus on alternate financing channels for data center companies. At the moment, we don't see evidence that ABS or CMBS supply has been displaced by corporate credit issuance but it is a possibility that as the cost of capital has come in cross many corners of the AI/DC build that there is not as much urgency to refinance into securitized. We think it is possible that supply this year and in 2027-2028 could lag our expectations if the credit markets – both public and private – continue to offer sufficient depth and competitive cost of capital for data center companies that are pursuing rapid growth. We are now also additionally considering the possibility that data center companies looking to go public could curtail debt issuance or use proceeds from those IPOs to pay down debt in order to de-lever in line with some of the existing public data center REITs.

Corporate credit investors have been wondering if there is sufficient depth of market in ABS and CMBS to absorb a potential wave of refinancings out of the high yield market. We think the structures we have seen thus far in credit offer things that securitized investors want, namely better transparency on tenants/leases and deleveraging, which could help demand and offset some of the downsides of those profiles (such as the crypto/neocloud dynamic, a risk not often seen in ABS and CMBS). The HY structures and credit profiles look very dissimilar to the ABS and CMBS assets today, which could temper demand as securitized credit investors work to come up to speed on yet another flavor of digital infra risk.

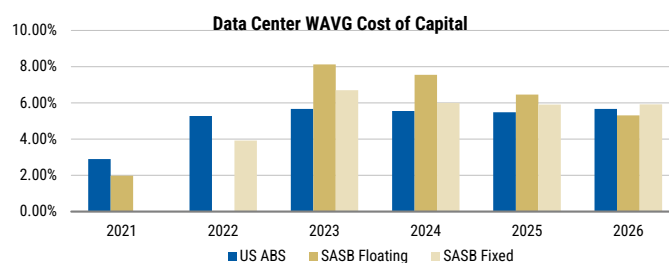
Elsewhere, we are seeing a pick-up in focus on chip/GPU financing and energy infrastructure. Investors continue to express an interest in shorter-dated, amortizing structures, which could help demand for chip-based financings as the underlying assets have a naturally shorter useful life than a data center. Securitized credit investors looking for risk akin to the tenant risk we describe above rather than asset or thematic risk may prefer these structures over data centers, or at least seek to diversify their digital infrastructure holdings and exposures using these newer debt instruments.

Exhibit 20: Gross issuance, data center securitized realized & forecast



Source: Intex, Bloomberg, CreditFlow, Trepp, Morgan Stanley Research

Exhibit 21: Data center ABS & CMBS WAVG cost of capital (at issuance)



Source: Intex, Bloomberg, CreditFlow, Trepp, Morgan Stanley Research. Data through 6/23/2026

Will securitized credit supply get displaced by the credit market?

2Q26 issuance in securitized credit has been slower than in 1Q26, with \$7.5 bn issued vs \$9.5bn in the prior quarter, or a decrease of ~20%, and with the bulk of Q2 issuance coming in its final weeks. YTD, supply is tracking at around \$16-17bn or just around halfway to our total issuance estimate of \$30-35bn. Until the latest deals priced towards the back half of June, and given the volume of data center and AI-related financing elsewhere in fixed income, securitized credit investors were left questioning whether the slowdown was related to the issuance in other markets, with securitized credit being displaced in preference for a market that offers greater depth of demand and more liquidity. We think it is too early to say that financing of data centers that might otherwise have been issued through ABS or CMBS will now pass through the HY or IG markets, but we are certainly looking for indications that issuers may gravitate towards markets where they can more easily execute in large size given their growth plans. Our view remains that companies will pursue a wide range of financing channels, and that issuance in the credit and loan markets will supplement, rather than replace, their ABS and CMBS financing needs.

Moreover, we think there is still a natural need for both the ABS and CMBS markets for two main reasons (and note that we have [written before](#) that from an asset perspective there is not much difference between the two).

First, the asset profile of data centers in HY and IG is largely dissimilar to those in credit, with the notable exception of QTS's Quest 10y bond. In [Exhibit 22](#) we provide a general profile of data centers found in both the corporate and securitized credit markets – while there are indeed some overlaps in asset types, with single hyperscaler campuses found in both, by and large the securitized credit market finances are relatively small, multi-tenant, multi-regional pools of data centers that are multi-purpose and operated by data center companies with years of experience, while the credit markets have thus far focused on single-tenant, single-campus AI data centers often being developed by relatively new players such as crypto companies and almost always with some construction risk. While we would not rule out multi-tenant, multi-region data centers backing corporate credit bonds, at this time we do think the focus of that market is really single-tenant, single-campus per deal, and expect it to remain that way for the time being.

Exhibit 22: Data center

Types of Assets Commonly Found by Market		
	Corporate Credit	Securitized Credit
Construction Risk	Vast Majority	None*
Workload	AI	AI, Cloud, Mixed Use
Number of Tenants	One	1-1000+
Number of Campuses	One	Often more than one
Size	50+ MWs	1-100MW

Source: Morgan Stanley Research. *One CMBS deal was issued pre-stabilization.

Second, data center companies that *are* issuing into the ABS and CMBS markets have lots of online capacity that does not yet back a publicly-issued deal. We estimate about 4GW of capacity sits across ABS shelves, and another 2GW in CMBS. Combined, though, we think these issuers have at least 20GW of operational capacity with another 25-30GW of capacity in their pipelines (estimates are very hard to arrive at given inconsistent reporting

and tracking of assets, but we think our estimates are in the right ballpark). While some financing of these assets is surely sitting elsewhere in ABF-style structures or in construction loans, only a fraction of that capacity needs to come through the securitized credit markets to see meaningful growth in issuance.

What other ABF/ABS-style deals are we likely to see, and how much?

The capex coming through fixed income markets is broadening out, and will likely continue to do so. The handful of chip financing deals we have seen thus far is just the start. While we continue to expect that the hyperscalers will have a preference to own rather than lease their compute, we do think that the growing neocloud ecosystem, the clear ability & willingness of the market to absorb these transactions with SPVs leasing out compute, and the involvement of Alphabet in the TPU/neocloud industry means there is real upside to the volume of deals that we could see this year and on a going-forward basis. **We also think that some of that risk could come through the securitized credit market either in late 2026 or by 2027.**

Elsewhere, investor focus on energy financing, specifically gas turbine ABF style structures is notable. We expect this to be an important emerging theme in 2H26, as more deals backed by energy infrastructure come through the private and potentially public markets. In addition, fiber ABS issuance is robust and has been well-bid by investors. Growth in the space has been fueled by enterprise players; networking infrastructure is obviously critical to the growth of AI data, and we expect to see continued supply along this theme over the coming months and years.

Disclosure Section

Mortgage Backed Securities (MBS) and Collateralized Mortgage Obligations (CMO)

Principal is returned on a monthly basis over the life of the security. Principal prepayment can significantly affect the monthly income stream and the maturity of any type of MBS, including standard MBS, CMOs and Lottery Bonds. Yields and average lives are estimated based on prepayment assumptions and are subject to change based on actual prepayment of the mortgages in the underlying pools. The level of predictability of an MBS/CMO's average life, and its market price, depends on the type of MBS/CMO class purchased and interest rate movements. In general, as interest rates fall, prepayment speeds are likely to increase, thus shortening the MBS/CMO's average life and likely causing its market price to rise. Conversely, as interest rates rise, prepayment speeds are likely to decrease, thus lengthening average life and likely causing the MBS/CMO's market price to fall. Some MBS/CMOs may have "original issue discount" (OID). OID occurs if the MBS/CMO's original issue price is below its stated redemption price at maturity, and results in "imputed interest" that must be reported annually for tax purposes, resulting in a tax liability even though interest was not received. Investors are urged to consult their tax advisors for more information. Government agency backing applies only to the face value of the CMO and not to any premium paid.

The information and opinions in Morgan Stanley Research were prepared by Morgan Stanley & Co. LLC, and/or Morgan Stanley C.T.V.M. S.A., and/or Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., and/or Morgan Stanley Canada Limited. As used in this disclosure section, "Morgan Stanley" includes Morgan Stanley & Co. LLC, Morgan Stanley C.T.V.M. S.A., Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., Morgan Stanley Canada Limited and their affiliates as necessary.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Carolyn L. Campbell; James Egan; Fernanda Lima; Vishwas Patkar; Vishwanath Tirupattur.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of June 30, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1544	42%	453	49%	29%	757	44%
Equal-weight/Hold	1577	43%	390	42%	25%	769	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	544	15%	89	10%	16%	204	12%
Total	3,668		933			1731	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures.

For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 14-9169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

The following authors are Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity securities: Carolyn L Campbell; Vishwanath Tirupattur; James Egan; Vishwas Patkar; Fernanda Lima.